

Cybersecurity Threat Analysis Report

Executive Summary

This report analyzes cybersecurity investigation data across three key areas: APT threat intelligence, malware analysis, and social engineering attack simulation. The investigation reveals a coordinated threat landscape involving Iranian APT groups, malicious zip files, and credential harvesting attempts.

1. APT Threat Intelligence Analysis

APT33 (Elfin) Profile

- **Threat Group:** APT33 (also known as Elfin)
- **Attribution:** Suspected Iranian threat group
- **Active Since:** At least 2013
- **MITRE ATT&CK ID:** G0064

Target Profile

- **Geographic Focus:** United States, Saudi Arabia, South Korea
- **Industry Sectors:** Aviation and Energy (primary focus)
- **Associated Groups:** HOLMIUM, Elfin, Peach Sandstorm

Intelligence Metadata

- **Created:** April 18, 2018
- **Last Modified:** April 11, 2024
- **Version:** 2.0
- **Contributors:** Dragos Threat Intelligence

2. Malware Analysis Results

Sample Information

- **File Hash:** e3c020cb7feda6ec304f3abf3ed71137842fe8ea984bfa218d0163eed9f49
- **File Name:** ppp.zip
- **File Size:** 496.82 KB
- **File Type:** Encrypted ZIP archive

VirusTotal Detection Results

- **Detection Rate:** 1/63 security vendors flagged as malicious
- **Community Score:** Low threat level indicated
- **Analysis Status:** Recently analyzed

Security Vendor Analysis

Vendor	Detection Status	Threat Classification
Microsoft	DETECTED	Trojan:Script/Wacatac.B!ml
AhnLab-V3	Undetected	–
AliCloud	Undetected	–
ALYac	Undetected	–
Antiy-AVL	Undetected	–
Acronis (Static ML)	Undetected	–
Alibaba	Undetected	–
Arcabit	Undetected	–

Key Findings

- **Password Protection:** File is password-protected, limiting deep analysis
- **Low Detection Rate:** Only Microsoft's engine detected malicious content
- **Threat Classification:** Script-based malware (Wacatac family)

3. Social Engineering Attack Simulation

Attack Vector Analysis

- **Tool Used:** Social Engineering Toolkit (SET) in Parrot OS
- **Target Platform:** Google Services (OAuth2 authentication)
- **Attack Type:** Credential harvesting via phishing

Technical Parameters Observed

```
GALX=SJLCkfqaqQM
OAuth2 URL: https://accounts.google.com/o/oauth2/auth
Service: ISO
DSH: 7381887106725792428
Background Response: JavaScript disabled
Post Message: 1
DN Connection: [Not specified]
Check Connection: [Active]
Checked Domains: youtube
```

Credential Harvest Results

- **Potential Username:** johnson (Email field)
- **Potential Password:** james (Password field)
- **Sign-in Status:** Sign-in attempt captured
- **Persistent Cookie:** Enabled
- **HTTP Response:** 302 redirect (successful form submission)

Attack Timeline

- **Initial Request:** 16/Jun/2025 18:12:38 (GET / HTTP/1.1 200)
- **Favicon Request:** 16/Jun/2025 18:12:39 (404 error)
- **Form Submission:** 16/Jun/2025 18:12:49 (POST /ServiceLoginAuth HTTP/1.1 302)

4. MITRE ATT&CK Framework Mapping

APT33 Tactics and Techniques

Initial Access

- **T1566:** Phishing (Email-based attacks)
- **T1190:** Exploit Public-Facing Application

Execution

- **T1059:** Command and Scripting Interpreter
- **T1204:** User Execution

Persistence

- **T1053:** Scheduled Task/Job
- **T1078:** Valid Accounts

Credential Access

- **T1110:** Brute Force
- **T1056:** Input Capture
- **T1555:** Credentials from Password Stores

Collection

- **T1005:** Data from Local System
- **T1039:** Data from Network Shared Drive

5. Risk Assessment

Threat Level: MEDIUM-HIGH

APT33 Risk Factors

- **Sophistication:** High (Nation-state actor)
- **Persistence:** High (Active since 2013)

- **Industry Impact:** Critical (Aviation/Energy sectors)

Malware Risk Factors

- **Detection Evasion:** High (Low AV detection rate)
- **Encryption:** File protection complicates analysis
- **Payload Unknown:** Encrypted content prevents full assessment

Social Engineering Risk Factors

- **Success Rate:** High (Credential capture successful)
- **Platform Targeting:** Google services (high user base)
- **Technical Sophistication:** Medium (OAuth2 exploitation)

6. Recommendations

Immediate Actions

1. **Enhanced Monitoring:** Implement detection rules for APT33 TTPs
2. **Email Security:** Strengthen anti-phishing controls
3. **User Training:** Conduct targeted awareness programs
4. **Credential Security:** Implement MFA for critical systems

Strategic Improvements

1. **Threat Intelligence:** Subscribe to APT33-specific intelligence feeds
2. **Behavioral Analysis:** Deploy advanced endpoint detection
3. **Network Segmentation:** Isolate critical aviation/energy systems
4. **Incident Response:** Update playbooks for Iranian APT activities

Technical Controls

1. **ZIP File Filtering:** Block password-protected archives
2. **Script Execution:** Restrict unauthorized script execution
3. **OAuth Monitoring:** Monitor for suspicious authentication flows
4. **Domain Reputation:** Implement real-time URL reputation checking

7. Conclusion

The analysis reveals a sophisticated threat landscape with Iranian APT33 activity, evasive malware distribution, and effective social engineering tactics. The low malware detection rate (1/63) and successful credential harvesting demonstrate the evolving nature of modern cyber threats. Organizations in the aviation and energy sectors should prioritize defensive measures against APT33's known tactics, techniques, and procedures.

The investigation highlights the importance of multi-layered security approaches, combining threat intelligence, technical controls, and user awareness to defend against advanced persistent threats and social engineering attacks.

Report Generated: June 16, 2025
Analysis Classification: TLP:WHITE
Next Review Date: July 16, 2025

ApplicationsPlacesSystem

ATT&CK Data & Tools | MITREAPT33, Group G0013 | VirusTotal - File - e3c020MalwareBazaar | DownloadAPT33, HOLMIUM, Elfin

https://attack.mitre.org/groups/G0064/

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ATT&CKcon 6.0 is coming October 14-15 in McLean, VA and live online. To potentially join us on stage, submit to our CFP by July 9th

GROUPS

APT33APT37APT38APT39APT41APT42APT5Aquatic PandaAxiomBackdoorDiplomacyBITTERBlackByteBlackOasisBlackTechBlue MockingbirdBouncing Golf

Home > Groups > APT33

APT33

APT33 is a suspected Iranian threat group that has carried out operations since at least 2013. The group has targeted organizations across multiple industries in the United States, Saudi Arabia, and South Korea, with a particular interest in the aviation and energy sectors.^{[1][2]}

ID: G0064

Associated Groups: HOLMIUM, Elfin, Peach Sandstorm

Contributors: Dragos Threat Intelligence

Version: 2.0

Created: 18 April 2018

Last Modified: 11 April 2024

Version Permalink

Associated Group Descriptions

Name	Description
HOLMIUM	[3]
Elfin	[4]

ApplicationsPlacesSystem

Parrot Terminal

File Edit View Search Terminal Help

[*] Information will be displayed to you as it arrives below:
192.168.1.222 - - [16/Jun/2025 18:12:38] "GET / HTTP/1.1" 200 - Privacy and Security Learning Resources
192.168.1.222 - - [16/Jun/2025 18:12:39] "GET /favicon.ico HTTP/1.1" 404 -
[*] WE GOT A HIT! Printing the output:
PARAM: GALX=SJLCKfgaqoM
PARAM: continue=https://accounts.google.com/o/oauth2/auth?zt=ChRsWFBwd2JmV1hIcDhtUdFldzBENhIfVWxsxSToNLW9MdThibw1TMfQzVUZFc1BBaURuWm1RSQxE2%88%99APsBz4gAAAAUy4_qd7Hbfz38w8kxnaNouLcRiD3YTjX
PARAM: service=lso
PARAM: dsh=-7381887106725792428
PARAM: _utf8=a
PARAM: bgsresponse=js_disabled
PARAM: pstMsg=1
PARAM: dnConn=
PARAM: checkConnection=
PARAM: checkedDomains=youtube
POSSIBLE USERNAME FIELD FOUND: Email=johnson
POSSIBLE PASSWORD FIELD FOUND: Passwd=james
PARAM: signIn=Sign+in
PARAM: PersistentCookie=yes
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.
192.168.1.222 - - [16/Jun/2025 18:12:49] "POST /ServiceLoginAuth HTTP/1.1" 302 -

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ATT&CK Data & Tools | M |APT30, Group G0013 | M |VirusTotal - File - e3c020cb7feda6ec304f3abf3ed7113784a2fe8eaa984bfa218d0163eeda9f49 | MalwareBazaar | Downlo...APT33, HOLMIUM, Elfina |

https://www.virustotal.com/gui/file/e3c020cb7feda6ec304f3abf3ed7113784a2fe8eaa984bfa218d0163eeda9f49/detection

Import bookmarks...Parrot OSHack The BoxOSINT ServicesVuln DBPrivacy and SecurityLearning Resources

e3c020cb7feda6ec304f3abf3ed7113784a2fe8eaa984bfa218d0163eeda9f49Sign inSign up

1/63Community Score

1/63 security vendor flagged this file as maliciousReanalyzeSimilarMore

e3c020cb7feda6ec304f3abf3ed7113784a2fe8eaa984bfa218d0163eeda9f49Size496.82 KBLast Analysis Datea moment agoZIPencrypted

DETECTIONDETAILSRELATIONSCOMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

This file is password-protected, security vendors may not have been able to look into it

Security vendors' analysisDo you want to automate checks?

Microsoft	Trojan:Script/Wacatac.B!ml	Acronis (Static ML)	Undetected
AhnLab-V3	Undetected	Alibaba	Undetected
AliCloud	Undetected	ALYac	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected

MenuVirusTotal - File - e3c...enterprise-attack-v17...Parrot Terminal

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